Power Requirements:

1. PZEM-004T (Metering Module)

- **Description**: Measures voltage, current, power, and energy consumption.
- **Operating Voltage**: 80-260V AC
- **Current Measurement**: Up to 100A
- **Power Consumption**: Approximately **0.5W**
- **Key Features**: High precision, real-time monitoring.

2. ESP32 WROOM 32 (Microcontroller Unit)

- **Description**: A versatile MCU with Wi-Fi and Bluetooth capabilities.
- **Operating Voltage**: 3.3V
- **Peak Current**: Up to **160mA** during Wi-Fi operation
- **Power Requirement**:

  $P = V \times I = 3.3\ V \times 0.16\ A =$ **0.528W**

- **Key Features**: Dual-core processor, integrated peripherals.

3. DS3231 (RTC Module)

- **Description**: Real-time clock with temperature compensation.
- **Operating Voltage**: 3.3V to 5V
- **Current Requirement**:
  - Battery mode: **0.1µA**
  - Active mode: **1-2mA**
- **Power Requirement** (worst case):

  $P = 5\ V \times 0.002\ A =$ **0.01 W**

4. AT24C256 (EEPROM Module)

- **Description**: 256K I2C EEPROM for data storage.
- **Operating Voltage**: 1.8V to 5.5V
- **Current Requirement**: Approximately **1mA**
- **Power Requirement**:

  $P = 5\ V \times 0.001\ A =$ **0.005 W**

5. 20x4 LCD (Liquid Crystal Display)

- **Description**: Displays information in a 20x4 character format.
- **Operating Voltage**: 5V
- **Current Requirement**: Approximately **2-20mA**
- **Power Requirement**:

  $P = 5\ V \times 0.02\ A =$ **0.1 W**

6. SIM800L (GPRS Module)

- **Description**: GSM/GPRS module for communication.
- **Operating Voltage**: 3.4V to 4.4V (recommended 4.0V)
- **Peak Current**: Up to **2A** during transmission
- **Power Requirement**:

  $P = 4\ V \times 2\ A =$ **8 W**

7. ZMPT101B (Voltage Sensor Module)

- **Description**: Measures AC voltage.
- **Operating Voltage**: 5V
- **Current Requirement**: Typically **10-20mA**
- **Power Requirement**:

  $P = 5\ V \times 0.02\ A =$ **0.1 W**

8. A1220 Hall Effect Sensor Module

- **Description**: Detects magnetic fields.
- **Operating Voltage**: 4.5V to 24V
- **Current Requirement**: Typically **10-15mA**
- **Power Requirement**:

  $P = 5\ V \times 0.015\ A =$ **0.075 W**

9. D2440-B (Solid State Relay)

- **Description**: Controls AC loads, ideal for high power applications.
- **Operating Voltage**: 5V DC control, 24-280V AC load
- **Control Current**: Typically **10-15mA**
- **Power Requirement**:

  $P = 5\ V \times 0.015\ A =$ **0.075 W**

10. TCA9548A I2C Multiplexer

- **Description**: An 8-channel I2C switch that allows multiple I2C devices to communicate on the same bus, managing communication between up to eight I2C devices with the same address.
- **Operating Voltage**: 1.65V to 5.5V
- **Current Requirement**: Typically **10mA**
- **Power Requirement**:

  P = 5 V × 0.01 A = **0.05 W**

Total Power Requirement Calculation:

$$\text{Total Power} = P_{\text{PZEM}} + P_{\text{ESP32}} + P_{\text{DS3231}} + P_{\text{AT24C256}} + P_{\text{LCD}} + P_{\text{SIM800L}} + P_{\text{ZMPT101B}} + P_{\text{A3144}} + P_{\text{D2440}} + P_{\text{TCA9548A}}$$

Substituting the values:

$$\text{Total Power} = 0.5 + 0.53 + 0.01 + 0.005 + 0.1 + 8 + 0.1 + 0.075 + 0.075 + 0.05 = 9.445\,\text{W}$$

Final Total Power:

The total power required for the system is **9.445W**.

Power Requirement Table:

Module	Voltage(V)	Current(Amp)	Power(W)
PZEM-004T	5	0.1	0.5
ESP32 WROOM 32	3.3	0.16	0.53
DS3231	5	0.002	0.01
AT24C256	5	0.001	0.005
20x4 LCD	5	0.02	0.1
SIM800L	4	2	8
ZMPT101B	5	0.02	0.1
A1220	5	0.015	0.075
D2440-B	5	0.015	0.075
TCA9548A	5	0.01	0.05
Total	-	-	9.445

On the basis of above power calculations of System we decided to use HLK-PM05 Hi-Link 5V 20W AC to DC Power Supply Module:

The **HLK-PM05** is a compact and efficient AC to DC power supply module that converts 85-265V AC input to a regulated 5V DC output. This module is designed to provide reliable power for various electronic applications, making it ideal for powering low-power devices such as microcontrollers, sensors, and communication modules. With a high power density and small form factor, it is suitable for integration into smart energy metering systems and other compact electronic devices.

Specifications

Specification	Details
Input Voltage	85-265V AC
Output Voltage	5V DC
Output Current	4A (max)
Output Power	20W (max)
Efficiency	> 82%
Ripple Voltage	< 50mV
Operating Temperature	-40°C to +85°C
Size	60mm x 36mm x 18mm
Weight	60g
Protection Features	Short Circuit, Overload, Over Temperature Protection